DSP · L08

Inverse z-transform

Cheat sheet · Day 15 · Sat 7 Nov · pack D8

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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INVERSE z (L08)  -- work in w = z^-1, factors as (1 - p w)
A_i = [(1 - p_i w) X]_{w = 1/p_i}      check: sum A_i (+quotient) = X at w=0
A/(1-p z^-1):  |z|>|p| -> A p^n u(n)        |z|<|p| -> -A p^n u(-n-1)
Double pole: 1/(1-pz^-1)^2 -> (n+1)p^n u(n)  |  -(n+1)p^n u(-n-1)
             find A2 by cover-up, then A1 from w = 0
Complex pair p = r e^{j th}:  2|A| r^n cos(th n + angle A) u(n)
                            = 2 r^n [ReA cos th n - ImA sin th n] u(n)
Improper (M >= N in z^-1): divide first; M = N -> constant b_N/a_N
   (ratio of HIGHEST z^-1 coeffs) -> constant*delta(n)
ROC: poles inside ROC's inner edge -> causal terms; outside outer edge -> anti-causal
N distinct pole radii -> N+1 ROCs -> N+1 signals. Stable = ROC contains |z|=1
Power series: coeff of z^-n = x(n). Causal: divide ascending z^-1.
   Left-sided: divide ascending z (highest z^-1 power first)
System ID: H = Y/X (factor, cancel); ROC from "causal"/"stable"; R_y >= R_h n R_x
LCCDE <-> H: y(n) + sum a_k y(n-k) = sum b_k x(n-k)  <->  H = B(z^-1)/(1 + sum a_k z^
   LCCDE alone does not fix the ROC
Causal <-> ROC outside outermost pole (incl. inf). Stable <-> ROC has |z|=1.
Causal+stable <-> all poles |p|<1. Anti-causal+stable <-> all |p|>1.
FIR (MA): M zeros, Mth-order pole at 0, always stable.
All-pole (AR): N poles, Nth-order zero at 0.
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